

Name: _____

Class: _____



Mr. Rawson • GCSE Computer Science

Year 11 2026

Programming Challenge: Arithmetic Operators

Modulus and Even/Odd Checker (HARD) — Answer Key

https://isaacomputerscience.org/questions/arithops_hard

- Write a Python program that asks the user for a number and checks whether it is even or odd.
- If the number is divisible by 2, print “Even”.
- Otherwise, print “Odd”.

Example Output:

```
> Enter a number: 7
```

```
Odd
```

Previous Knowledge Needed

- Input converted with `int()` (line 2)
- Remainder with `%` (line 9)
- IF and ELSE (line 15)

Alfred's answer, v2.

```
1 # asks the user for a number = int(input())
2 number = int(input("Enter a number: "))
3
4 # divisible by 2 = no remainder after dividing by 2
5 # % gives the remainder: 7 % 2 = 1, 8 % 2 = 0
6 # so number % 2 == 0 means divisible by 2
7 # the OCR exam writes % as MOD
8 # If the number is divisible by 2 = if number % 2 == 0:
9 if number % 2 == 0:
10     # print "Even" = print("Even")
11     print("Even")
12
13 # Otherwise = else:
14 # else takes every number the if did not: the ones with remainder 1
15 else:
16     # print "Odd" = print("Odd")
17     print("Odd")
```